SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: Computer Vision with OpenCV COURSE CODE: DSA02

COURSE OUTCOME COVERED

CO3: *Analyze shape representation methods and techniques such as contours, regions, deformable models, and multi-resolution analysis. (BL4)*

Assessment Tool Weightage: 30%

QUESTIONS: 2 Total Marks:30 Annexure A

Case study

Code of Conduct:

I K. Pujitha (192524149) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: *K. Pujitha*

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understandi ng of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly	
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts	
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning	

Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided	
Clarity	10	Clear, well structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand	

Annexure A

Case study

Case Study 1: Defect Detection in Industrial Parts

A manufacturing company wants to automatically detect defects in gears and machine components using computer vision. The images of the components are binary and may contain small protrusions or missing parts.

Questions:

A. How can region-based representation and medial axis computation help in identifying defective parts?

B. Discuss how noise in image acquisition could affect shape analysis and propose preprocessing steps to improve accuracy.

C. Compare the effectiveness of Fourier descriptors versus wavelet descriptors for classifying normal vs. defective components.

Case Study 2: Medical Image Segmentation

A hospital is developing a system to segment organs from CT scans. The organ boundaries are irregular and may vary slightly between patients.

Questions:

A. Explain how active contours (snakes) can be applied to segment the organ boundaries.

B. Discuss the challenges of initializing contours and tuning parameters for accurate segmentation.

C. How could level set methods improve segmentation for cases where organs split or merge in the images?

Case Study 1: Defect Detection in Industrial Parts

A. Region-Based Representation and Medial Axis Computation

The objective is to automatically distinguish normal industrial parts from defective parts such as gears and machine components. Since the input images are binary, the component can be separated from the background and represented as a connected region.

1. Region-Based Representation

Region-based representation analyzes the entire area occupied by the object.

For a gear or machine component, the following properties can be extracted:

- **Area** – detects missing or additional material.
- **Centroid** – identifies changes in the object's mass distribution.
- **Perimeter** – detects irregular boundaries.
- **Shape and connected components** – identify missing portions or unwanted regions.

For example, if a gear has a missing tooth, its area and perimeter will differ from those of a normal gear. Similarly, an unwanted protrusion increases the region area and changes the boundary.

Therefore, region properties can be compared with a reference model of a normal component to identify defects.

2. Medial Axis Computation

The medial axis or skeleton reduces the object to a thin representation while preserving its structural information.

It is useful because:

- It represents the central structure of the component.
- It preserves important topological information.
- Branches and endpoints can reveal structural abnormalities.
- The skeleton of a defective part can be compared with the skeleton of a normal part.

For example, a missing gear tooth can cause a change in the skeleton structure, while an unwanted protrusion can introduce an abnormal branch.

Analysis

Thus, region representation detects changes in the overall shape and area, whereas medial axis analysis detects structural changes. Using both provides more reliable defect detection than using either method alone.

B. Effect of Noise and Preprocessing

Noise during image acquisition can significantly affect shape analysis. Camera sensors, poor lighting, dust, vibration, and transmission errors can introduce unwanted pixels.

Effects of Noise

Noise can:

- Create false protrusions.
- Produce small isolated regions.
- Fill small gaps or holes.
- Change area and perimeter measurements.
- Distort the object's boundary.
- Produce unwanted branches during skeletonization.

For example, a single noisy pixel near the edge of a gear may appear as a small protrusion and could be incorrectly classified as a manufacturing defect.

Proposed Preprocessing Pipeline

Binary Image → Noise Removal → Morphological Processing → Connected Component Filtering → Boundary Refinement → Shape Analysis

Steps

1. Median Filtering

Removes salt-and-pepper noise while preserving important edges.

2. Thresholding

Ensures that the component and background are properly separated.

3. Morphological Opening

Removes small unwanted objects and noise.

4. Morphological Closing

Fills small gaps and holes in the component.

5. Connected Component Analysis

Small isolated regions that are not part of the actual component can be removed.

6. Boundary Smoothing

Reduces small irregularities before extracting shape descriptors.

7. Skeletonization

The medial axis is calculated only after the image has been cleaned.

Analysis

Preprocessing is essential because even small noise can create false defects. A clean binary image produces more accurate area, perimeter, boundary, medial axis, Fourier descriptors, and wavelet descriptors.

C. Fourier Descriptors vs. Wavelet Descriptors

Both descriptors can represent the shape of an industrial component, but they have different strengths.

Criteria	Fourier Descriptors	Wavelet Descriptors
Representation	Frequency-based boundary representation	Multi-resolution representation
Global shape	Excellent	Good
Local defects	Moderate	Excellent
Small protrusions	May be missed	Can detect effectively
Missing tooth	Good	Very good
Noise handling	Requires filtering	Better localization of noise

Computational complexity	Lower	Higher
Best application	Overall shape classification	Localized defect detection

Fourier Descriptors

Fourier descriptors represent the object's boundary using frequency components.

For a normal gear, the boundary produces a characteristic set of coefficients. A defective gear changes these coefficients.

Advantages:

- Compact representation.
- Effective for overall shape comparison.
- Can be normalized for translation, rotation, and scale.
- Computationally efficient.

Limitation: Very small local defects may not be represented well when only a few low-frequency coefficients are retained.

Wavelet Descriptors

Wavelet descriptors analyze the boundary at different scales and positions.

This makes them particularly useful for detecting localized defects such as:

- Small protrusions.
- Missing teeth.
- Small cracks or irregularities.
- Local boundary distortions.

Advantages:

- Captures both global and local shape information.
- Provides multi-resolution analysis.
- Better suited for small localized defects.

Limitation: It is more complex and computationally expensive than Fourier analysis.

Overall Solution Design

A reliable industrial defect detection system can combine both approaches:

Input Binary Image
↓
Noise Removal
↓
Morphological Processing
↓
Connected Component Extraction
↓
Region Feature Extraction
↓
Medial Axis/Skeleton Computation
↓
Fourier + Wavelet Descriptors
↓
Compare with Normal Reference
↓
Normal / Defective Classification

Why this solution is effective

- Region features detect changes in area and overall shape.
- Medial axis detects structural abnormalities.
- Fourier descriptors capture global boundary differences.
- Wavelet descriptors identify small localized defects.
- Preprocessing prevents noise from being incorrectly classified as a defect.

Case Study 2: Medical Image Segmentation

The following answer is structured to match the **Level 4 – Exemplary** rubric: clear understanding, correct application of concepts, detailed analysis, feasible solution design, and clarity.

A. Application of Active Contours (Snakes)

The objective is to accurately segment an organ from a **CT scan**, where the organ boundaries may be irregular and vary between patients.

1. Active Contour Concept

An **active contour or snake** is a deformable curve that moves toward the actual boundary of an object.

The contour is represented as:

$$C(s)=[x(s),y(s)]$$

It is controlled by an energy function:

$$E=E_{internal}+E_{external}$$

where:

- **Internal energy** controls the smoothness and continuity of the contour.
- **External energy** attracts the contour toward image features such as edges and intensity changes.

The general objective is:

$$C^{*}=\arg\min_{C} E(C)$$

The contour is initialized near the organ and iteratively moves until it reaches the organ boundary.

2. Application to CT Organ Segmentation

The process can be:

CT Image → Preprocessing → Initial Contour → Energy Minimization → Boundary Detection → Segmented Organ

1. The CT image is first enhanced or denoised.
2. An initial contour is placed approximately around the organ.
3. The snake is allowed to deform.
4. Image edges and intensity differences attract the contour toward the organ boundary.
5. Internal forces prevent excessive deformation and maintain smoothness.
6. The process continues until the contour stabilizes around the organ.

Analysis

Active contours are suitable because organ boundaries are not always perfectly circular or regular. The contour can adapt its shape to irregular boundaries, making it more flexible than simple fixed-shape segmentation methods.

B. Challenges in Initialization and Parameter Tuning

Accurate segmentation depends strongly on the **initial contour and model parameters**.

1. Initialization Challenges

The initial contour must usually be placed reasonably close to the organ.

Problems occur when:

- The contour is placed too far from the organ.
- The initial contour is incorrectly shaped.
- Multiple organs have similar intensity.
- The organ boundary is weak or unclear.
- Noise creates false edges.

If the contour starts far away, it may converge to the wrong boundary.

2. Parameter Tuning

The important parameters include:

- **Internal/smoothness parameter** – controls how smooth the contour remains.
- **Edge attraction parameter** – controls the influence of image edges.
- **Balloon or inflation force** – controls whether the contour expands or contracts.
- **Iteration count** – determines how long the contour evolves.

Effect of Incorrect Parameters

Parameter Problem	Possible Effect
Too much smoothness	Important organ boundary details may be lost
Too little smoothness	Contour becomes irregular due to noise
Weak edge force	Contour may not reach the organ boundary
Strong edge force	Contour may stop at false edges
Excessive iterations	Contour may move beyond the correct boundary

Too few iterations	Incomplete segmentation

Proposed Solution

A practical solution is to use:

Image preprocessing + approximate automatic initialization + adaptive parameter selection + stopping criteria

For example, image denoising can reduce false edges, while an approximate organ location can provide a better starting contour.

Analysis

Therefore, initialization and parameter selection are major factors affecting segmentation accuracy. A well-initialized contour with appropriately balanced forces can produce a much more accurate organ boundary.

C. Level Set Methods for Splitting and Merging Organs

A major limitation of traditional active contours is that they represent the boundary as an explicit curve. When an object splits into multiple regions or two regions merge, handling the topology becomes difficult.

Level set methods solve this problem by representing the contour implicitly as a level set function:

$\phi(x,y,t)$

The organ boundary is represented by the zero level:

$\phi(x,y,t)=0$

The contour evolves according to a level set evolution equation.

Advantages for Medical Images

Level set methods can naturally handle topological changes.

Case 1: Splitting

If one contour needs to separate into two parts, the level set can automatically split into multiple contours.

Example:
If an organ region contains two connected anatomical structures, the evolving level set can separate them without manually creating two contours.

Case 2: Merging

If two contours need to join into one region, the level set can automatically merge them.

This is difficult for conventional explicit snakes because the curve representation may require special handling when its topology changes.

Comparison

Feature	Active Contours (Snakes)	Level Set Methods
Boundary representation	Explicit curve	Implicit function
Irregular boundaries	Good	Very good
Initialization sensitivity	High	Moderate
Splitting	Difficult	Natural
Merging	Difficult	Natural
Topological changes	Limited	Excellent
Computational cost	Lower	Generally higher

Overall Solution Design

A robust CT organ segmentation system can use the following pipeline:

CT Scan
↓
Noise Reduction & Contrast Enhancement
↓
Initial Organ Localization

↓
Initial Contour
↓
Active Contour / Level Set Evolution
↓
Boundary Refinement
↓
Segmented Organ
↓
Accuracy Evaluation

Why this solution is effective

- Active contours adapt to irregular organ boundaries.
- Preprocessing reduces noise and false edges.
- Good initialization improves convergence.
- Parameter tuning balances smoothness and boundary attraction.
- Level set methods handle splitting and merging automatically.
- The combination provides a more flexible solution for patient-to-patient anatomical variation.